

PRATEEK SINGH

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GSSoC 2025 Contributor | Hacktoberfest 2025 Contributor | CodeSocial 2025 Contributor

EDUCATION

Indian Institute of Information Technology, Guwahati

B.Tech in Computer Science & Engineering | Graduation: 2027

TECHNICAL SKILLS

- Programming Languages: Java, Python, C++, JavaScript, TypeScript, HTML, CSS
- Frontend: React.js, TailwindCSS, Vite, CSS Modules, Context API
- Backend: Node.js, Express.js, Servlet/JSP, REST APIs
- Databases: MySQL, MongoDB, Mongoose, Firebase (Auth & Firestore)
- AI/ML: PyTorch, Scikit-learn, GANs (Pix2Pix / U-Net / PatchGAN), OpenRouter API (DeepSeek)
- Tools: Git, GitHub, VS Code, Linux, Jupyter Notebook, Multer, node-cron, Vercel, JDBC
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, Albumentations
- Core Skills: Competitive Programming, Problem Solving, Debugging, OOP, Secure Coding (OWASP)
- Coursework: DSA, Operating Systems, DBMS, AI, Computer Architecture, Discrete Mathematics, Optimization Techniques

PROJECTS

AuraSpot - AI-Powered Real Estate Marketplace | [GitHub](#)

React, TypeScript, Node.js, Express.js, MongoDB, Firebase, DeepSeek AI

- Full-stack real estate platform where property owners can list and manage properties, and tenants/buyers can search and connect with them.
- Used DeepSeek AI (via OpenRouter) to score properties, flag suspicious listings, suggest fair rent prices, and match users to relevant properties.
- Built real-time chat between owners and tenants, rent agreements, monthly payment tracking, and automated reminders for due/overdue payments.
- Added a maintenance request module where tenants can raise issues with priority levels and track status from submission to resolution.
- Owner dashboard shows rent collection trends, occupancy rates, and maintenance stats over time.
- Login via Email/Password or Google. Owners get trust badges (Verified, Trusted, Top Seller) based on activity and ratings.

Sketch to Colorization using Conditional GAN | [GitHub](#)

Python, PyTorch, U-Net, PatchGAN, Albumentations, Jupyter Notebook

- Built a cGAN model (based on Pix2Pix) that takes anime-style black and white sketches and outputs colored versions.
- Generator uses a U-Net architecture with skip connections so fine sketch details don't get lost during the coloring process.
- Discriminator is a PatchGAN that checks small patches of the image instead of the whole thing, which helps produce sharper and more realistic textures.
- Trained on the Anime Sketch Colorization Pair dataset. The entire workflow - data loading, training, and inference - is in one notebook.

Music Popularity Prediction using Machine Learning | [GitHub](#)

Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

- ML model that predicts whether a song will be popular based on audio features like tempo, energy, loudness, and danceability.

- Covered the full workflow from scratch: data cleaning, exploratory analysis, feature selection, model training, and evaluation.
- Tried multiple classification models and compared them using Accuracy, Precision, and Recall. Included charts to make the results easier to read.

Eventrix - Event Management System | [GitHub](#)

Java, MySQL, JDBC, Servlet/JSP, HTML/CSS

- Web app for managing events with separate dashboards for users and admins.
- Handles login, event creation, updates, and deletion with proper role-based access control.
- Backend follows the DAO pattern to keep database logic separate and the code easier to maintain.

Study Buddy - AI-Powered Chrome Extension | [GitHub](#)

JavaScript, Node.js, OpenRouter API, HTML/CSS

- Chrome extension that helps students by answering academic questions using multiple LLMs via OpenRouter.
- Handles token limits carefully to stay within free-tier usage, and picks the best available model dynamically based on the query.
- Clean popup UI with a Node.js backend that processes requests and returns responses in real time.

Mini WebApp Collection - Notes, Weather, To-Do List, Text-to-Speech | [To-Do List](#) [Weather App](#) [Text-to-Speech](#) [Notes App](#)

HTML, CSS, JavaScript, OpenWeatherMap API

- A set of small browser-based apps built to practice DOM manipulation, browser APIs, and responsive UI design.

EXPERIENCE

Open Source Contributor - GSSoC 2025, Hacktoberfest 2025 & CodeSocial 2025

- Raised and merged multiple PRs across different repos - mostly bug fixes, code cleanup, and documentation updates.
- Reviewed code for security issues and followed OWASP guidelines when making changes.
- Worked with maintainers and contributors from different time zones using Git, GitHub Issues, and PR reviews.
- Got comfortable with real open source workflows - branching, rebasing, CI checks, and writing good commit messages.